

机器学习基础作业6

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1. 使用前向概率和后向概率推导

$$P(\mathbf{O}|\lambda) = \sum_{l=1}^N \sum_{k=1}^N \alpha_t(l) a_{lk} b_k(O_{t+1}) \beta_{t+1}(k) \quad (1)$$

证明. 其中前向概率

$$\alpha_t(l) = P(O_1, O_2, \dots, O_t, q_t = s_l | \lambda) \quad (2)$$

后向概率

$$\beta_{t+1}(k) = P(O_{t+2}, O_{t+3}, \dots, O_T | q_{t+1} = s_k, \lambda) \quad (3)$$

状态转移概率

$$a_{lk} = P(q_{t+1} = s_k | q_t = s_l) \quad (4)$$

观测发射概率

$$b_k(O_{t+1}) = P(O_{t+1} | q_{t+1} = s_k) \quad (5)$$

考虑全概率公式

$$P(\mathbf{O}|\lambda) = \sum_{l=1}^N \sum_{k=1}^N P(O_1, \dots, O_T, q_t = s_l, q_{t+1} = s_k | \lambda) \quad (6)$$

注意到

$$P(O_1, \dots, O_t, q_t = s_l | \lambda) = \alpha_t(l) \quad (7)$$

$$P(q_{t+1} = s_k | q_t = s_l) = a_{lk} \quad (8)$$

$$P(O_{t+1} | q_{t+1} = s_k) = b_k(O_{t+1}) \quad (9)$$

$$P(O_{t+2}, \dots, O_T | q_{t+1} = s_k) = \beta_{t+1}(k) \quad (10)$$

从而

$$P(O_1, \dots, O_T, q_t = s_l, q_{t+1} = s_k | \lambda) = \alpha_t(l) a_{lk} b_k(O_{t+1}) \beta_{t+1}(k) \quad (11)$$

从而

$$P(\mathbf{O}|\lambda) = \sum_{l=1}^N \sum_{k=1}^N \alpha_t(l) a_{lk} b_k(O_{t+1}) \beta_{t+1}(k) \quad (12)$$

□

2. 证明维特比算法中 δ 的递推公式

证明. 假设隐马尔可夫模型 $\lambda = (A, B, \pi)$

状态集合 $Q = \{q_1, q_2, \dots, q_N\}$, 状态序列 $I = (i_1, \dots, i_T)$, 观测集合 $V = \{v_1, \dots, v_M\}$, 观测序列 $O = (o_1, \dots, o_T)$, 状态转移矩阵 $A = [a_{ij}]$ 其中 $a_{ij} = P(i_{t+1} = q_j | i_t = q_i)$, 观测概率矩阵 $B = [b_j(k)]$, 其中 $b_j(k) = P(o_t = v_k | i_t = q_j)$, 初始状态概率向量 $\pi = [\pi_i]$

定义

$$\delta_t(i) = \max_{i_1, i_2, \dots, i_{t-1}} P(i_1, i_2, \dots, i_{t-1}, i_t = q_i, o_1, o_2, \dots, o_t | \lambda) \quad (13)$$

对于

$$\delta_{t+1}(j) = \max_{i_1, \dots, i_t} P(i_1, \dots, i_t, i_{t+1} = q_j, o_1, \dots, o_{t+1} | \lambda) \quad (14)$$

展开联合概率

$$P(i_1, \dots, i_{t+1}, o_1, \dots, o_{t+1}) = P(i_1, \dots, i_t, o_1, \dots, o_t) P(i_{t+1} | i_t) P(o_{t+1} | i_{t+1}) \quad (15)$$

代入公式

$$\delta_{t+1}(j) = \max_{i_1, \dots, i_t} [P(i_1, \dots, i_t, o_1, \dots, o_t | \lambda) a_{i_t j} b_j(o_{t+1})] \quad (16)$$

注意到 $b_j(o_{t+1})$ 与之前的状态 $i_1, \dots, i_t$ 无关, $a_{i_t j}$ 只与 i_t 有关, 从而

$$\delta_{t+1}(j) = \max_{1 \leq i \leq N} [(\max_{i_1, \dots, i_{t-1}} P(i_1, \dots, i_{t-1}, i_t = q_i, o_1, \dots, o_t | \lambda))] b_k(o_{t+1}) \quad (17)$$

注意到

$$\max_{i_1, \dots, i_{t-1}} P(i_1, \dots, i_{t-1}, i_t = q_i, o_1, \dots, o_t | \lambda) = \delta_t(i) \quad (18)$$

从而

$$\delta_{t+1}(j) = \max_{1 \leq i \leq N} [\delta_t(i) a_{ij}] b_j(o_{t+1}) \quad (19)$$

从而

$$\begin{aligned} \delta_1(i) &= \pi_i b_i(o_1), i = 1, 2, \dots, N \\ \delta_t(i) &= \max_{1 \leq j \leq N} [\delta_{t-1}(j) a_{ji}] b_i(o_t), i = 1, 2, \dots, N \\ P^* &= \max_{1 \leq i \leq N} \delta_T(i) \end{aligned} \quad (20)$$

这也体现了动态规划思想: t 时刻最优路径一定是由 $t-1$ 的某个最优子路径扩展而来的。 $\square$

3. 计算观测概率

$$\begin{aligned} q_1 : b_1(v_1) &= 0.2, b_1(v_2) = 0.5, b_1(v_3) = 0.3 \\ b_2(v_1) &= 0.6, b_2(v_2) = 0.2, b_2(v_3) = 0.2 \\ b_3(v_1) &= 0.3, b_3(v_2) = 0.3, b_3(v_3) = 0.4 \end{aligned} \quad (21)$$

下面定义两个局部变量

$\delta_t(i)$, 时刻 t 处于状态 q_i 的所有可能路径中概率最大的一条路径的概率。

$\psi_t(i)$, 时刻 t 处于状态 q_i 的概率最大路径中, 时刻 $t-1$ 所处的状态。

初始化: $t = 1$ 时刻

观测值 $O_1 = v_3$, 计算各状态的初始局部概率

$$\delta_1 = \pi^T \circ b(v_3) = [0.18, 0.04, 0.08], \psi_1 = [0, 0, 0]$$

$t = 2$ 时刻，观测值 $O_2 = v_2$ ，中间转移矩阵 M_2

$$M_2 = \text{diag}(\delta_1) \cdot A = \begin{bmatrix} 0.18 & 0 & 0 \\ 0 & 0.04 & 0 \\ 0 & 0 & 0.08 \end{bmatrix} \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.108 & 0.036 & 0.036 \\ 0.012 & 0.020 & 0.008 \\ 0.024 & 0.024 & 0.032 \end{bmatrix} \quad (22)$$

从而 $\text{max_col}(M_2) = [0.108, 0.036, 0.036]$, $\psi_2 = [1, 1, 1]$

$$\delta_2 = \text{max_col}(M_2) \circ b(v_2) = [0.108, 0.036, 0.036] \circ [0.5, 0.2, 0.3] = [0.054, 0.0072, 0.0108] \quad (23)$$

$t = 3$ 时刻，观测值 $O_3 = v_1$ ，中间转移矩阵 M_3

$$M_3 = \text{diag} A = \begin{bmatrix} 0.054 & 0 & 0 \\ 0 & 0.0072 & 0 \\ 0 & 0 & 0.0108 \end{bmatrix} \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.0324 & 0.0108 & 0.0108 \\ 0.00216 & 0.0036 & 0.00144 \\ 0.00324 & 0.00324 & 0.00432 \end{bmatrix} \quad (24)$$

从而 $\text{max_col}(M_3) = [0.0324, 0.0108, 0.0108]$, $\psi_3 = [1, 1, 1]$

$$\delta_3 = \text{max_col}(M_3) \circ b(v_1) = [0.0324, 0.0108, 0.0108] \circ [0.2, 0.6, 0.3] = [0.00648, 0.00648, 0.00324] \quad (25)$$

最大值 $P^* = 0.00648$ 出现在状态 q_1, q_2 ，从而有两条最优路径： $(q_1, q_1, q_1), (q_1, q_1, q_2)$